

WEEK 1 REPORT

Foundations of Mechanical Design

1. Introduction

Mechanical design is an important field of mechanical engineering that involves the development of components, products, machines, and systems that perform specific functions safely, efficiently, and reliably. The mechanical design process involves understanding the requirements, selecting suitable materials, creating concepts, developing CAD models, preparing engineering drawings, applying appropriate dimensions and tolerances, and considering manufacturing and assembly requirements.

During Week 1 of my Mechanical Design Internship, I focused on understanding the fundamental concepts of mechanical design and becoming familiar with the tools and practices used by mechanical design engineers. The main areas covered during this week were **Mechanical Design fundamentals, CAD modelling using SolidWorks, and the basics of Geometric Dimensioning and Tolerancing (GD&T).**

2. Fundamentals of Mechanical Design

Mechanical design is not limited to creating a 3D model. A good design should satisfy functional, dimensional, manufacturing, safety, and economic requirements.

Some important factors considered during mechanical design are:

- Function and performance of the component
- Material selection
- Strength and durability
- Dimensions and tolerances
- Manufacturing process
- Ease of assembly
- Safety and reliability
- Cost and material availability

A typical mechanical design workflow can be represented as:

Requirement → Concept Development → Design Calculation → CAD Modelling → Engineering Drawing → Manufacturing → Inspection → Testing

Understanding this workflow helped me recognize how CAD modelling is connected to the overall product development process.

3. Introduction to SolidWorks

SolidWorks is a Computer-Aided Design (CAD) software widely used for developing mechanical components and assemblies. It allows engineers and designers to create accurate 2D sketches, 3D models, assemblies, and engineering drawings.

During this week, I became familiar with the basic SolidWorks interface and explored the **Part Design** environment. I studied the process of creating a sketch on a selected plane and converting the sketch into a three-dimensional component.

Some basic features studied include:

- Creating a new Part
- Selecting Front, Top, and Right planes
- Creating 2D sketches
- Applying sketch dimensions
- Applying geometric relations
- Extruded Boss/Base
- Extruded Cut
- Fillet
- Chamfer
- Basic viewing and model manipulation

I also understood the importance of maintaining proper dimensions and relations in a sketch so that the final model remains accurate and easy to modify.

4. Understanding GD&T

Geometric Dimensioning and Tolerancing (GD&T) is a system used to communicate allowable variations in the geometry of a component on engineering drawings.

In manufacturing, it is practically impossible to produce every component with absolutely zero dimensional variation. Therefore, tolerances are specified to define acceptable limits.

I studied the basic purpose of GD&T and its importance in manufacturing and inspection. I was introduced to concepts such as:

- Datum
- Form
- Orientation
- Location
- Profile
- Runout
- Dimensional tolerance

I understood that GD&T helps designers communicate design requirements clearly to manufacturers and inspectors. It also helps ensure that components can be manufactured and assembled correctly while maintaining the required functionality.

5. Design Considerations

While studying the fundamentals of mechanical design, I understood that a component should not only look correct in a CAD model but should also be practical to manufacture and use.

For example, while designing a component, the designer should consider:

Material → Manufacturing Process → Dimensions → Tolerances → Assembly → Function → Inspection

The selection of material and manufacturing process can affect the dimensions, surface finish, strength, cost, and overall performance of a component. Therefore, these factors need to be considered during the design stage.

6. Software and Working Environment Setup

As part of the Week 1 task, I set up and explored the required design environment. I opened SolidWorks and became familiar with its interface and basic modelling workflow. I also practiced creating a basic sketch and developing a simple 3D model using standard modelling features.

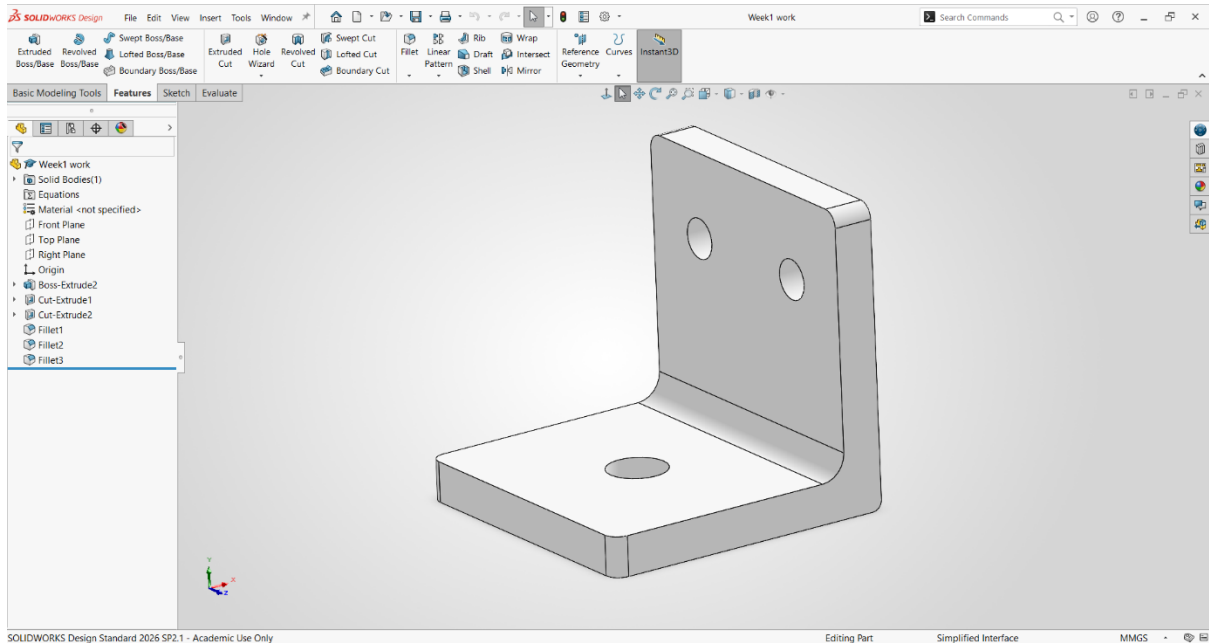


Figure 1: Flanged Mounting Bracket

7. Learning Outcomes

At the end of Week 1, I developed a basic understanding of the mechanical design process and the role of CAD software in product development. I became familiar with the fundamental SolidWorks workflow, including sketch creation, dimensions, geometric relations, and basic 3D modelling features.

I also gained an introductory understanding of GD&T and learned why dimensions, tolerances, and datums are important for manufacturing and inspection.

Overall, this week provided a strong foundation for the upcoming internship tasks. The knowledge gained will help me work more effectively on **3D modelling, assemblies, engineering drawings, and mechanical design projects** in the following weeks.

Conclusion

Week 1 helped me understand that mechanical design combines engineering principles, CAD modelling, manufacturing considerations, and technical communication. The practical exposure to SolidWorks along with the introduction to GD&T has given me a better understanding of how a mechanical component progresses from an initial requirement to a manufacturable design. I look forward to applying these fundamentals to more practical design tasks during the upcoming weeks of the internship.